

Rates of Change (1.3)

Today's Target



"I can calculate any average rate of change and any rate of change. I can interpret any rate calculated. I can apply what I have learned in familiar and unfamiliar settings."

A1. Rates of Change



Recall:

slope of tangent to $y = f(x)$ at $x = a$

average rate of change

(simply stated)

The change in the dependent variable per change in the independent variable.

Average Velocity and Velocity

An object moving in a straight line is an example of a rate of change model. Simply stated, its velocity is its speed at a point in time, in a given direction. Thus, average velocity is its average speed over a course of time, in a given direction. An axis represents the line of motion. In this section of the course we generally choose either a horizontal axis (x) or vertical axis (y). Moreover, movement to the right (or up) is considered a positive direction, whereas movement to the left (or down) is considered the negative direction.

Definitions

Displacement is the position of an object, in a given direction.

Average Velocity is the average rate of change in displacement per unit of time, in a given direction.

Velocity is the rate of change in displacement per unit of time, in a given direction.

Formula

Let the position s of an object at time t be denoted as $s(t)$. Thus,

$$\text{Average Velocity} = \frac{s(t_2) - s(t_1)}{t_2 - t_1}$$

With your groups

Activity A firework is projected vertically upward. It's position above is given by

$$s(t) = -5t^2 + 30t$$

Where s is the height in meters, and t is the time in seconds.

- a) Find the average velocity during the fourth second.
- b) Find the average velocity from 3 seconds to 3.5 seconds.
- c) Outline an approach that will estimate the velocity at exactly 3 seconds.

You need not perform calculations.

$$a) \quad S(t) = -5t^2 + 30t$$

$$v_{avg} = \frac{S(3) - S(4)}{3 - 4}$$

$$v_{avg} = \frac{45 - 40}{-1}$$

$$v_{avg} = -5 \text{ m/s}$$

@ t = 3

$$S(3) = -5(3)^2 + 30(3)$$

$$S(3) = 45$$

$$S(4) = -5(4)^2 + 30(4) = -80 + 120$$

$$S(4) = 40$$

$$b) \quad v_{avg} = \frac{S(3.5) - S(3)}{3.5 - 3}$$

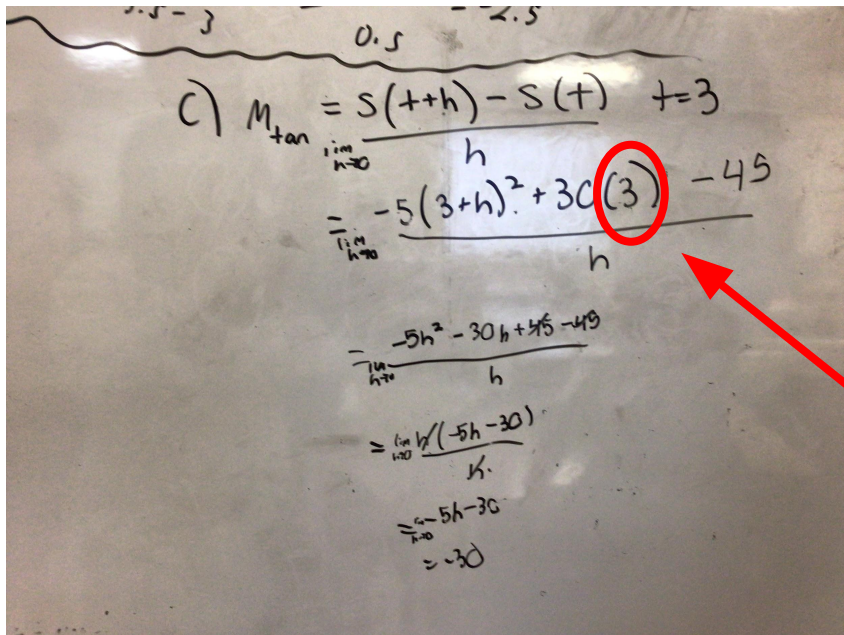
$$= \frac{43.75 - 45}{0.5}$$

$$= -2.5 \text{ m/s}$$

Example 1

Calculate the velocity of the firework at 3 seconds in Example 1.

$$s(t) = -5t^2 + 30t$$

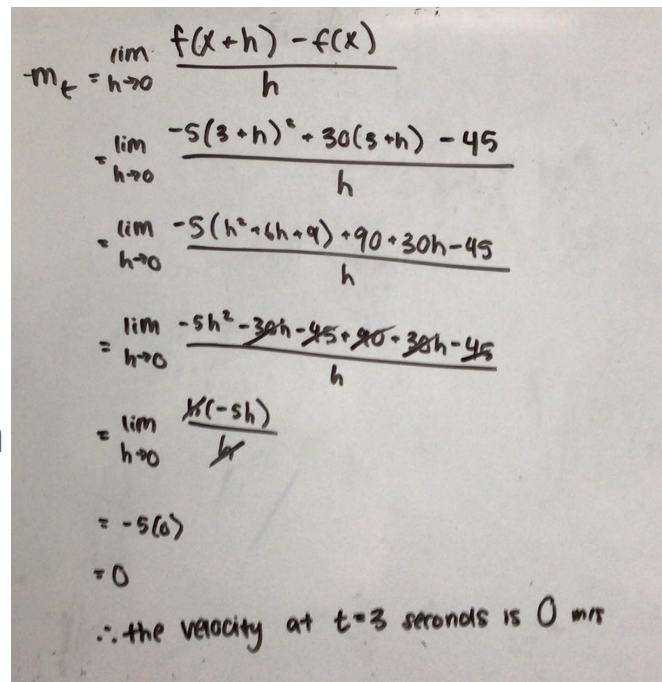


Handwritten student work for Example 1. The student is calculating the limit of the difference quotient for the position function $s(t) = -5t^2 + 30t$ at $t=3$. The work shows the following steps:

$$\begin{aligned} c) \quad m_{\tan} &= \lim_{h \rightarrow 0} \frac{s(3+h) - s(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5(3+h)^2 + 30(3) - 45}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5h^2 - 30h + 45 - 45}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5h^2 - 30h}{h} \\ &= \lim_{h \rightarrow 0} (-5h - 30) \\ &= -30 \end{aligned}$$

A red circle highlights the term $30(3)$ in the numerator of the second line, and a red arrow points from the text "Substitution mistake" to it.

Substitution
mistake
 $3+h$ instead



Handwritten student work for Example 1, showing the correct solution. The student is calculating the limit of the difference quotient for the position function $s(t) = -5t^2 + 30t$ at $t=3$. The work shows the following steps:

$$\begin{aligned} m_t &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5(3+h)^2 + 30(3+h) - 45}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5(h^2 + 6h + 9) + 90 + 30h - 45}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5h^2 - 30h - 45 + 90 + 30h - 45}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5h^2 - 30h}{h} \\ &= \lim_{h \rightarrow 0} (-5h - 30) \\ &= -30 \end{aligned}$$

The final conclusion is: $\therefore$ the velocity at $t=3$ seconds is 0 m/s .

Average Rate of Change and Rate of Change

average
rate of change
(formally stated)

$$m_{\text{sec}} = v_{\text{avg}} = \frac{s(t_2) - s(t_1)}{t_2 - t_1}$$

rate of change
(formally stated)

$$m_t = v = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t}$$

"Go the distance" with these:

Page 29...#	2*	3	10	14**	15c	20***	22b
<i>Level</i>	2	3	3	3	2	4+	4+

*interpret also using the term velocity

**for 14c, graph on paper instead of using a graphing calc. Also the final answer should be $0 < x < 80$

*** get rid of the word "approximate". I want the EXACT rate of change